

Complexity of Randomization Setup in Platform Trials - A Statistician's Perspective

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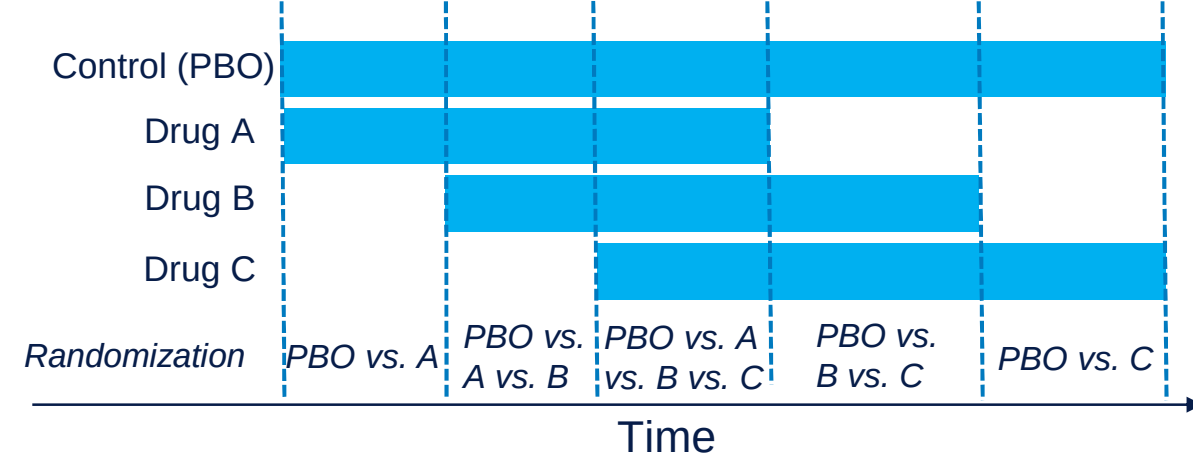
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Overview

- 1 Background and Motivation
- 2 Randomization Strategy
 - 2.1 Schedule Setup
 - 2.2 Allocation Ratio
- 3 Discussion

Background: Platform Trial



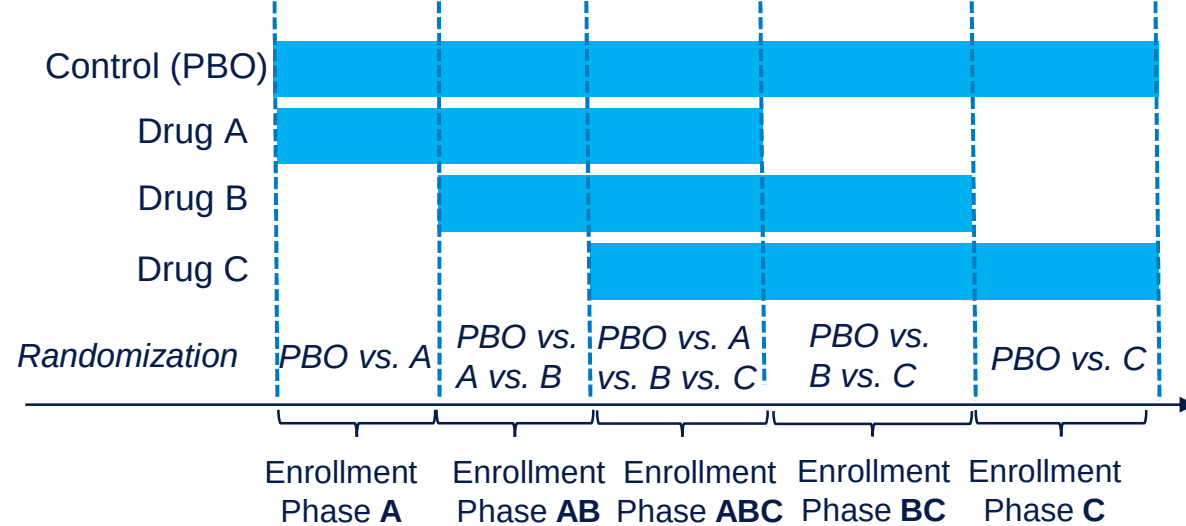
Accelerate Timeline & Reduce Operation Costs

- Evaluating multiple therapies simultaneously for a disease
- Enabling the dynamic addition or discontinuation of arms
- Utilizing a single master protocol and shared infrastructure

Reduce Overall Sample Size

- Using a common reference therapy arm for benchmarking all experimental therapies
- Offering an innovative design strategy for clinical trials to improve statistical efficiency under a limited total sample size

Motivation: Design Considerations for Randomization (RN) Setup

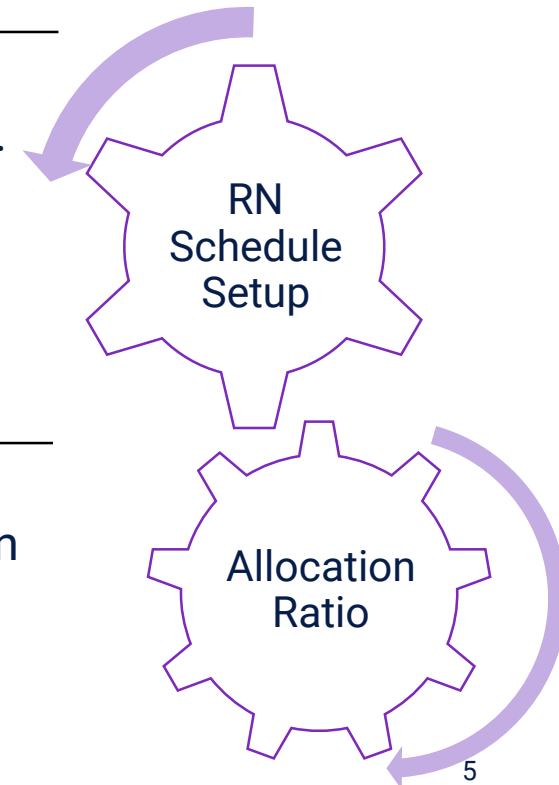


Distinct Eligibility Criteria Across Arms

- With multiple drugs actively enrolling, distinct eligibility criteria across arms (e.g., pts ineligible to one arm if w/ prior failure to the same MOA) prevent a single RN schedule.
- Potential re-enrollment: allowing subjects to re-enter the trial for a different drug (“recycling”) adds complexity to the RN schedule setup.

Dynamic Randomization Ratio

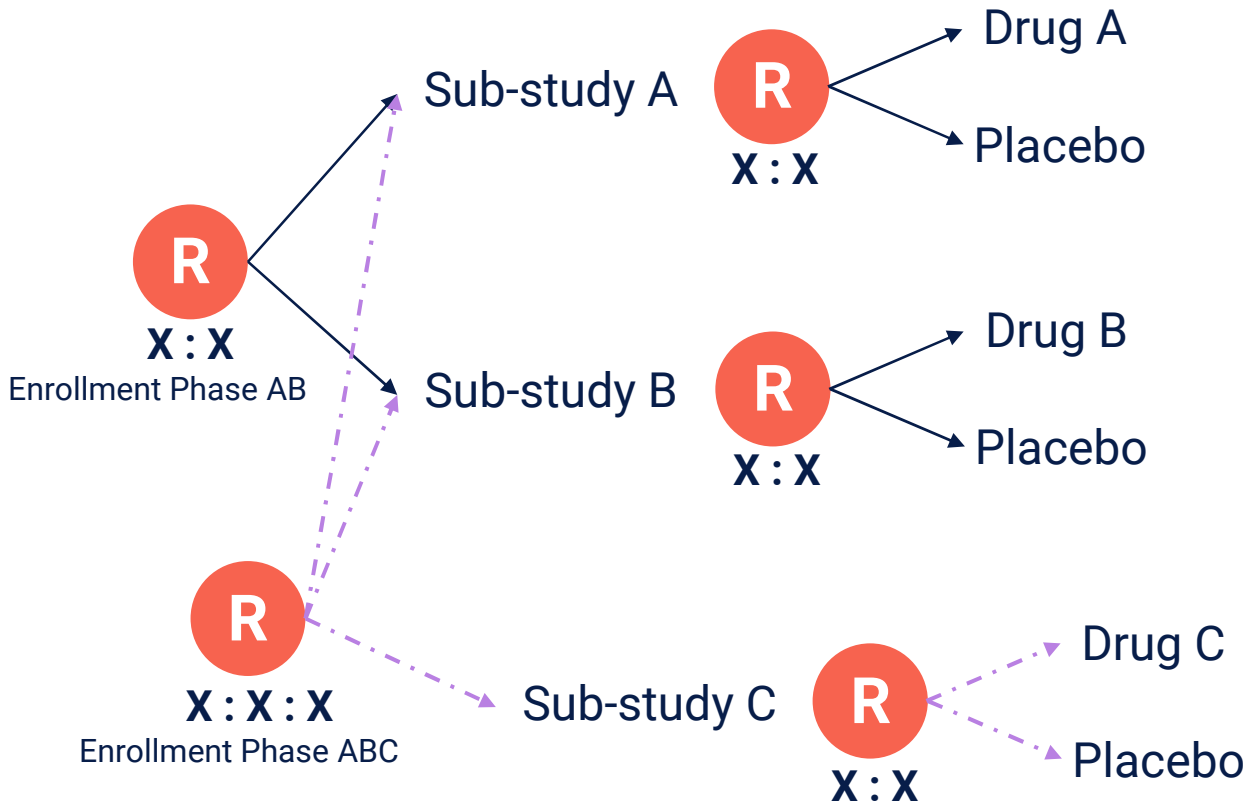
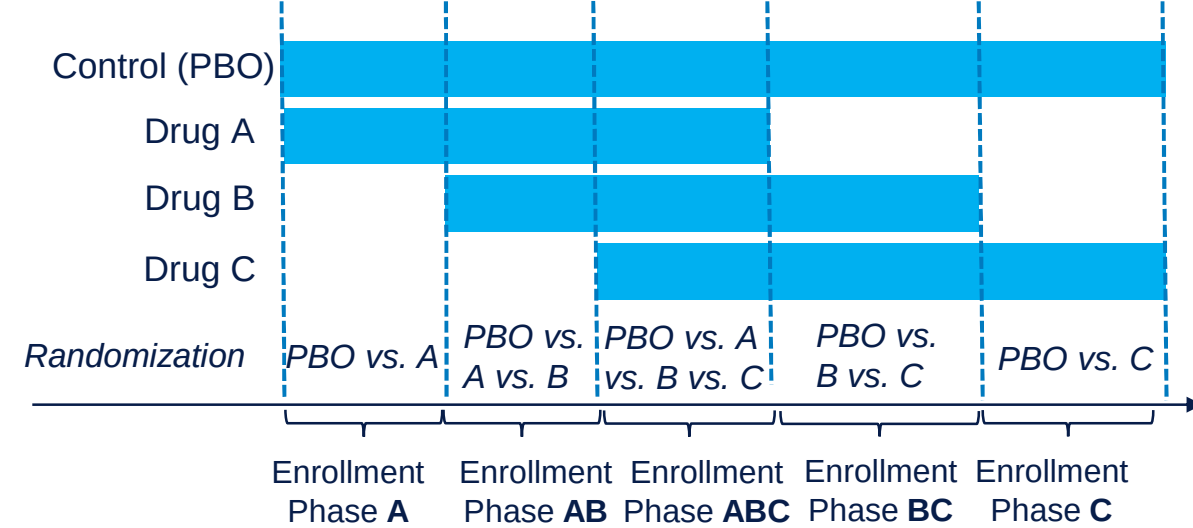
- When evaluating multiple drugs against a common control, FDA guidance suggests allocating more subjects to the shared control arm and adjusting dynamically based on the # of active treatment arms to increase overall power within a fixed total sample size. Following this suggestion, the RN ratio may change across enrollment phases.



RN Schedule Strategy 1:

Two-Step RN with Updates

Upon Arm Addition or Removal



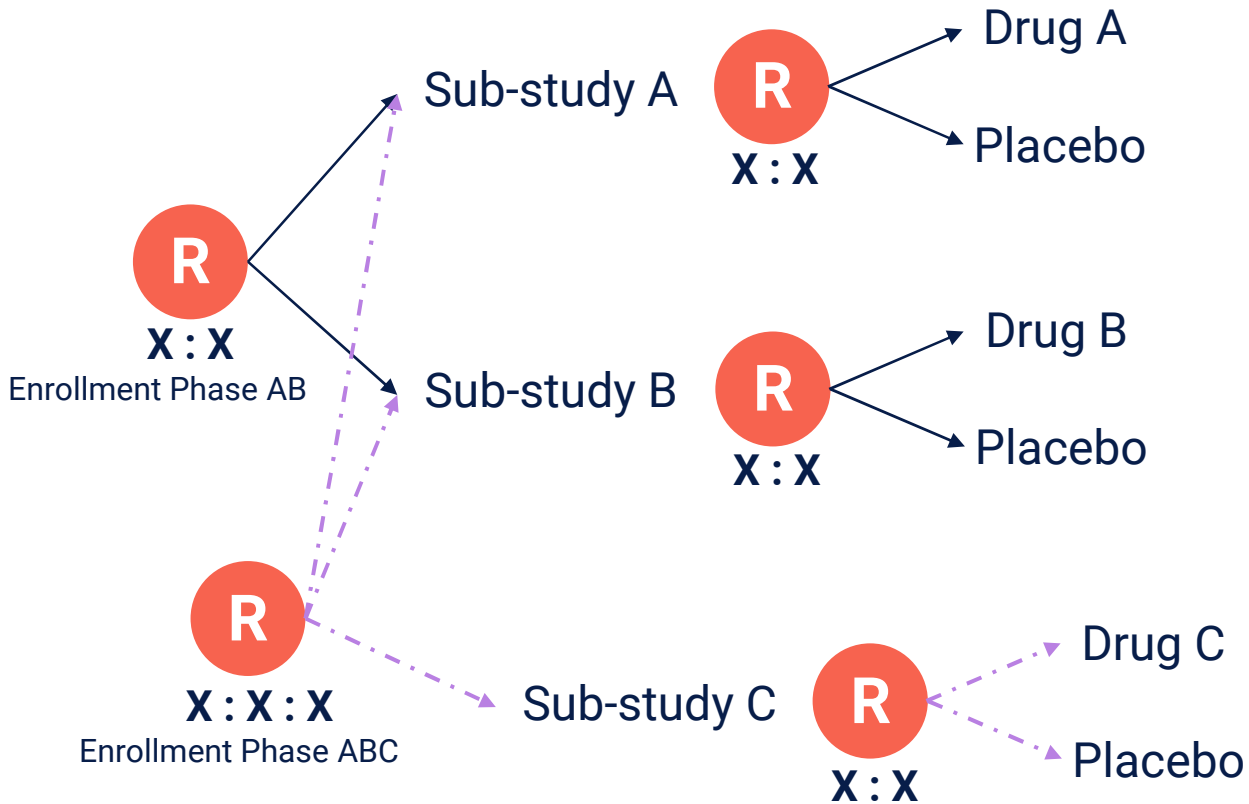
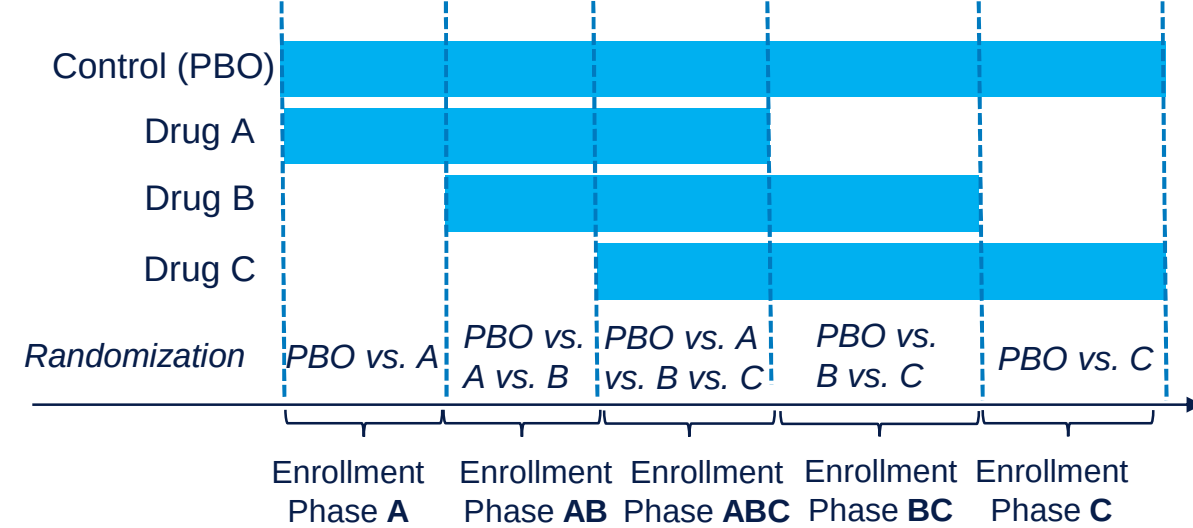
Under Two-Step RN, *the platform trial is decomposed into multiple sub-studies*, allowing each investigational drug to be evaluated independently while leveraging common trial resources.

- Step 1 RN: assign subjects to a sub-study
- Step 2 RN: randomize subjects within that sub-study

RN Schedule Strategy 1:

Two-Step RN with Updates

Upon Arm Addition or Removal



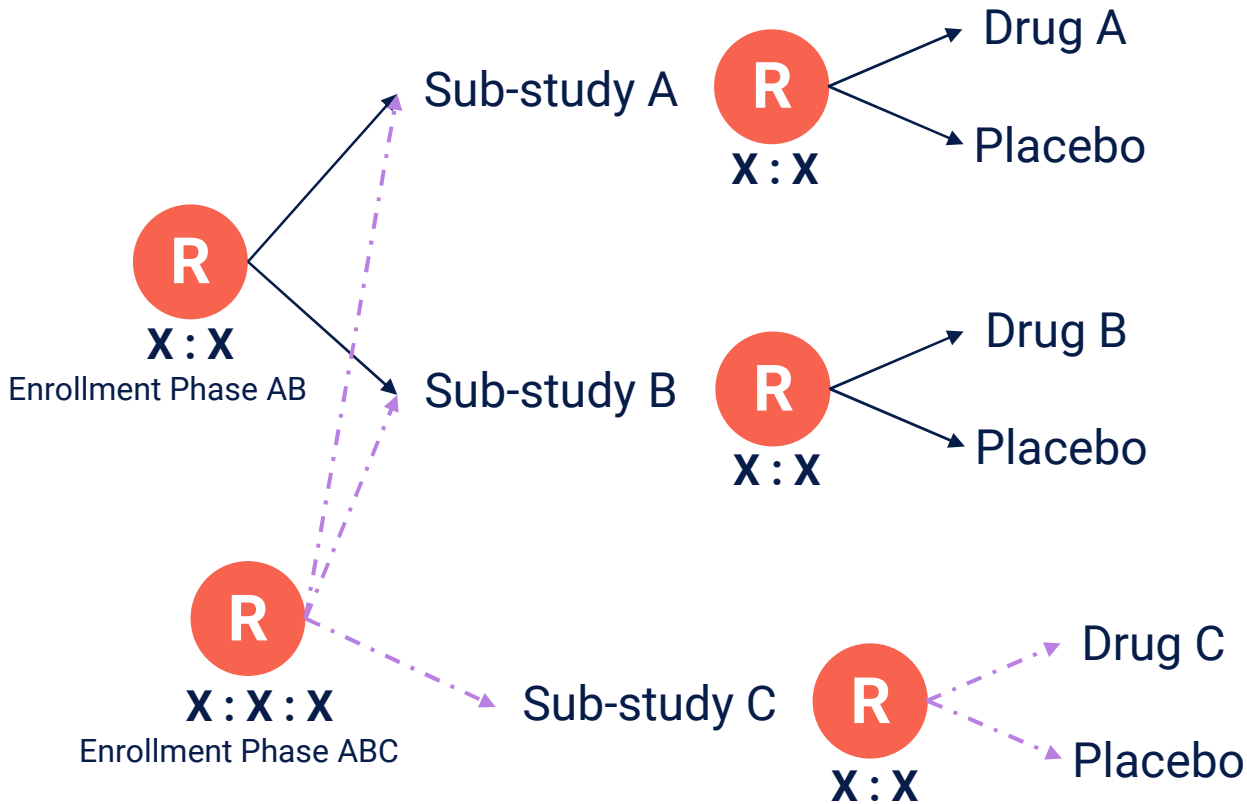
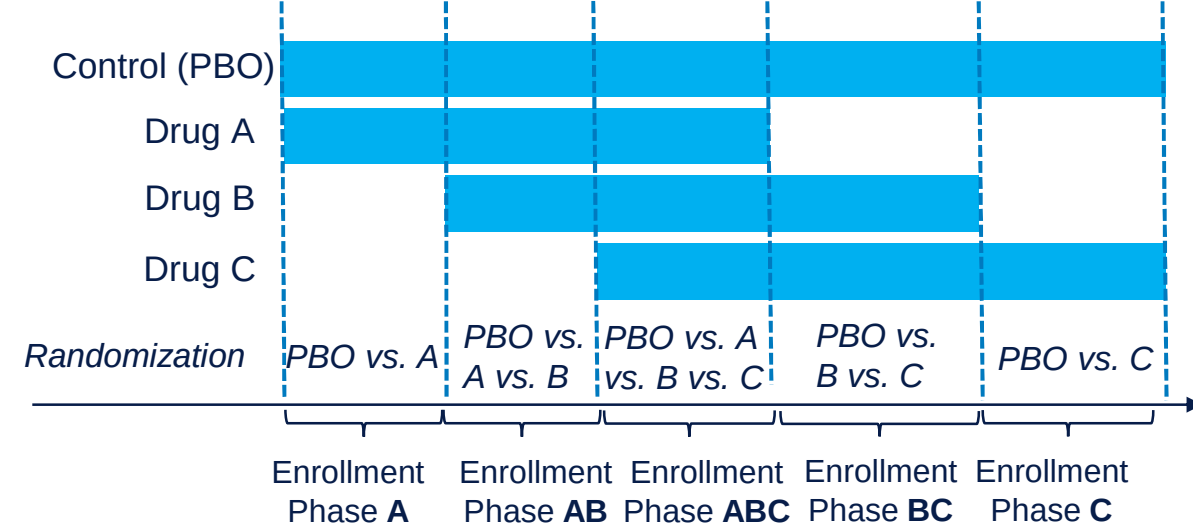
This strategy accommodates

- distinct eligibility criteria across arms by predefining rules at Step 1 RN. E.g., subjects NOT eligible for Drug B can ONLY be assigned to Sub-study A or C (when available)
- dynamic randomization ratio by updating RN ratios upon updates of RN schedule in case of arm addition or removal
- partial blinding may be considered esp. in case of different administration routes (e.g., IV/SC)

RN Schedule Strategy 1:

Two-Step RN with Updates

Upon Arm Addition or Removal

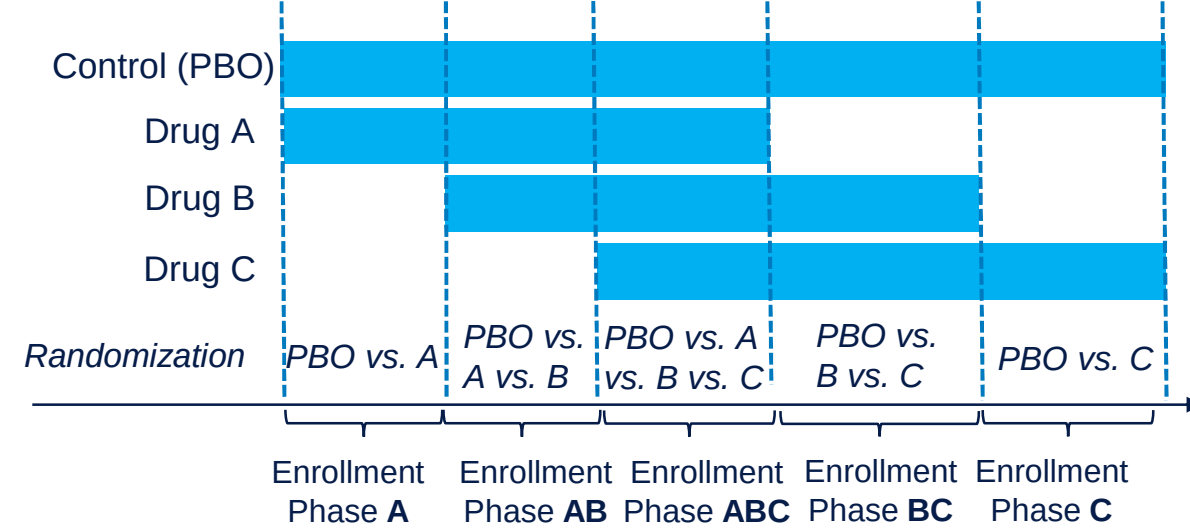
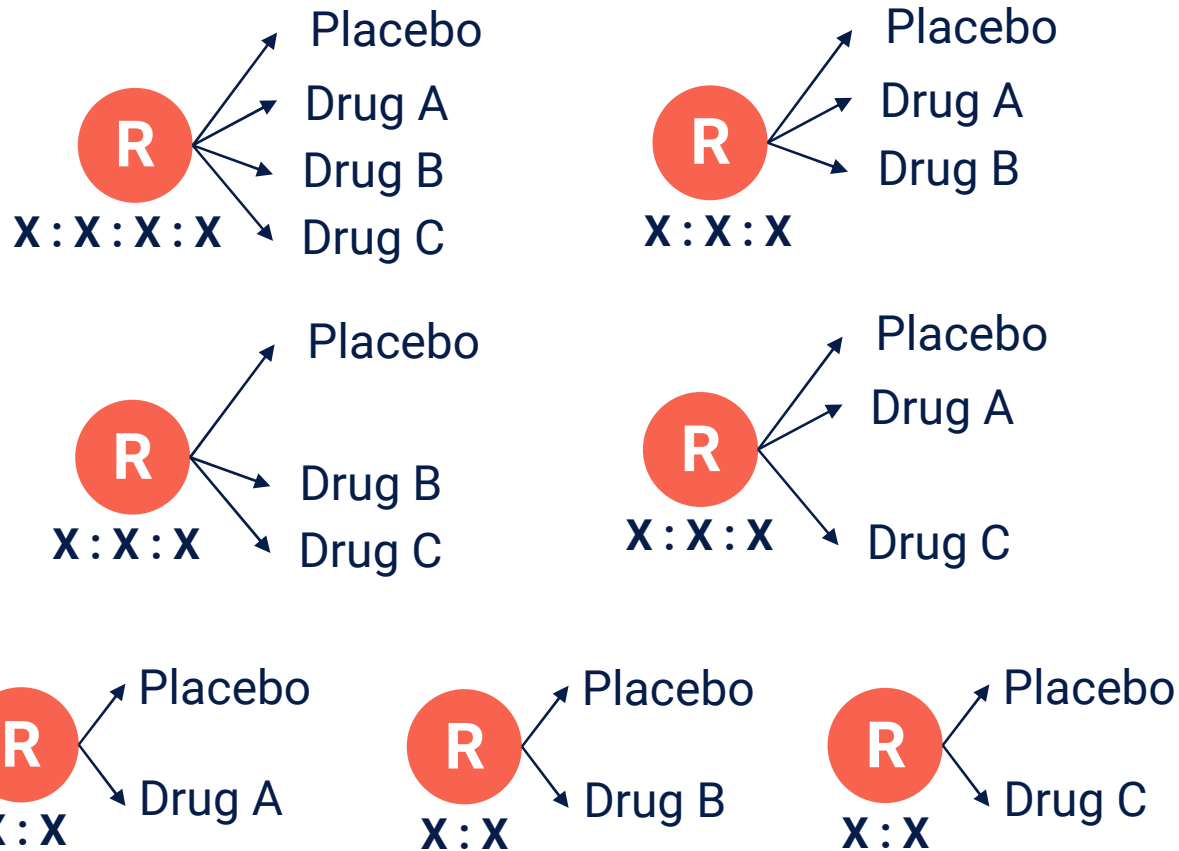


Note of inferential impact:

- further justification is needed for *pooling placebo data*
- *Stratification*, when needed, is preferred at both steps or at least at Step 2

RN Schedule Strategy 2:

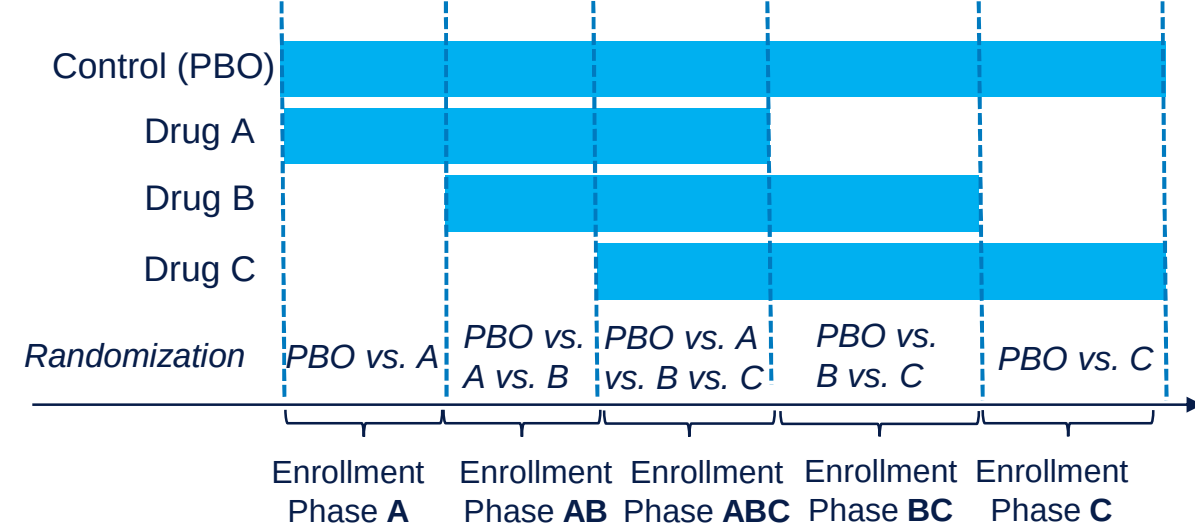
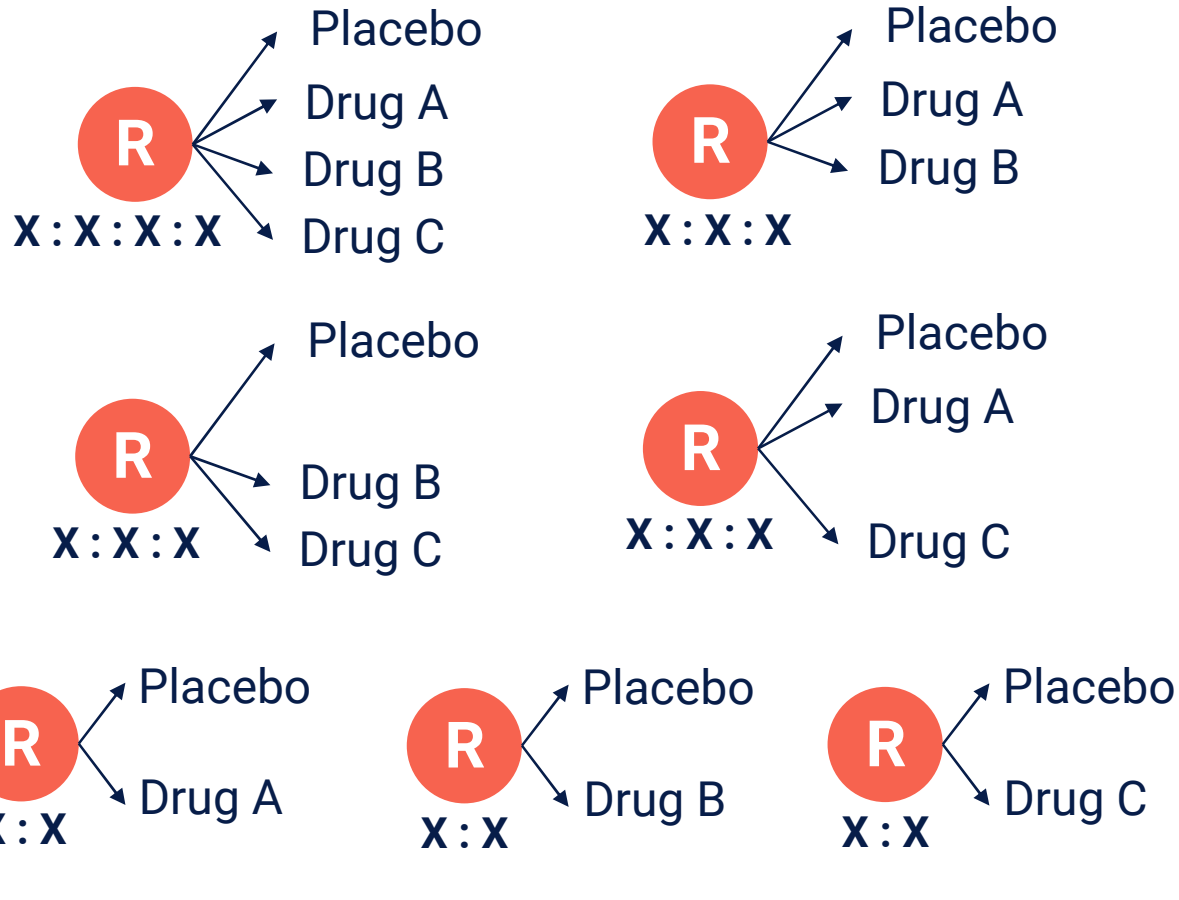
One-Step RN Sets



One-Step RN Sets are constructed to enumerate possible treatment-eligibility combinations; for example, with 3 investigational drugs, the RN sets include schedules for

- the all-drug sets + PBO
- any 2-drug sets + PBO
- any 1-drug sets + PBO

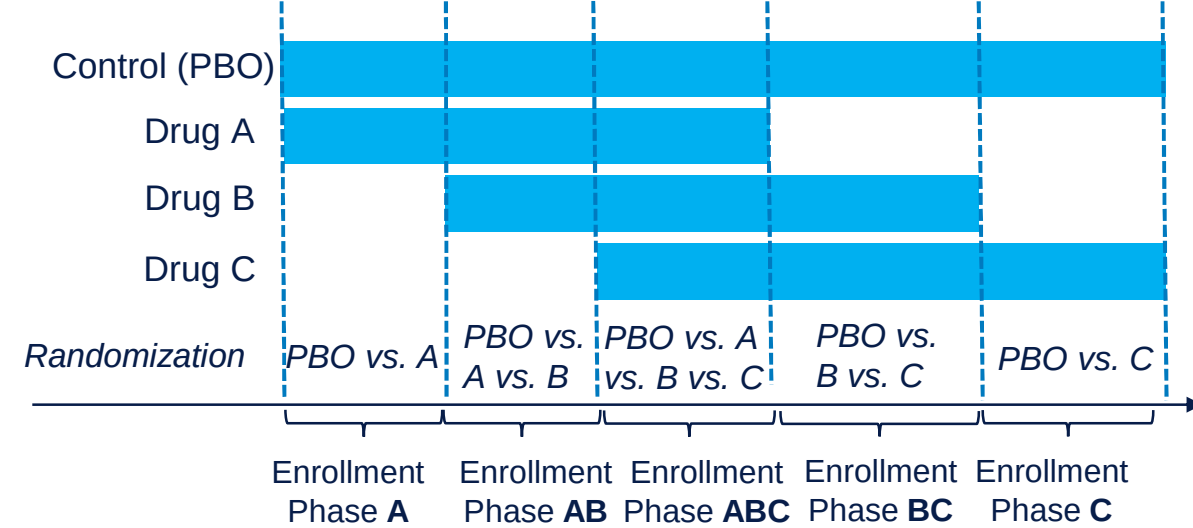
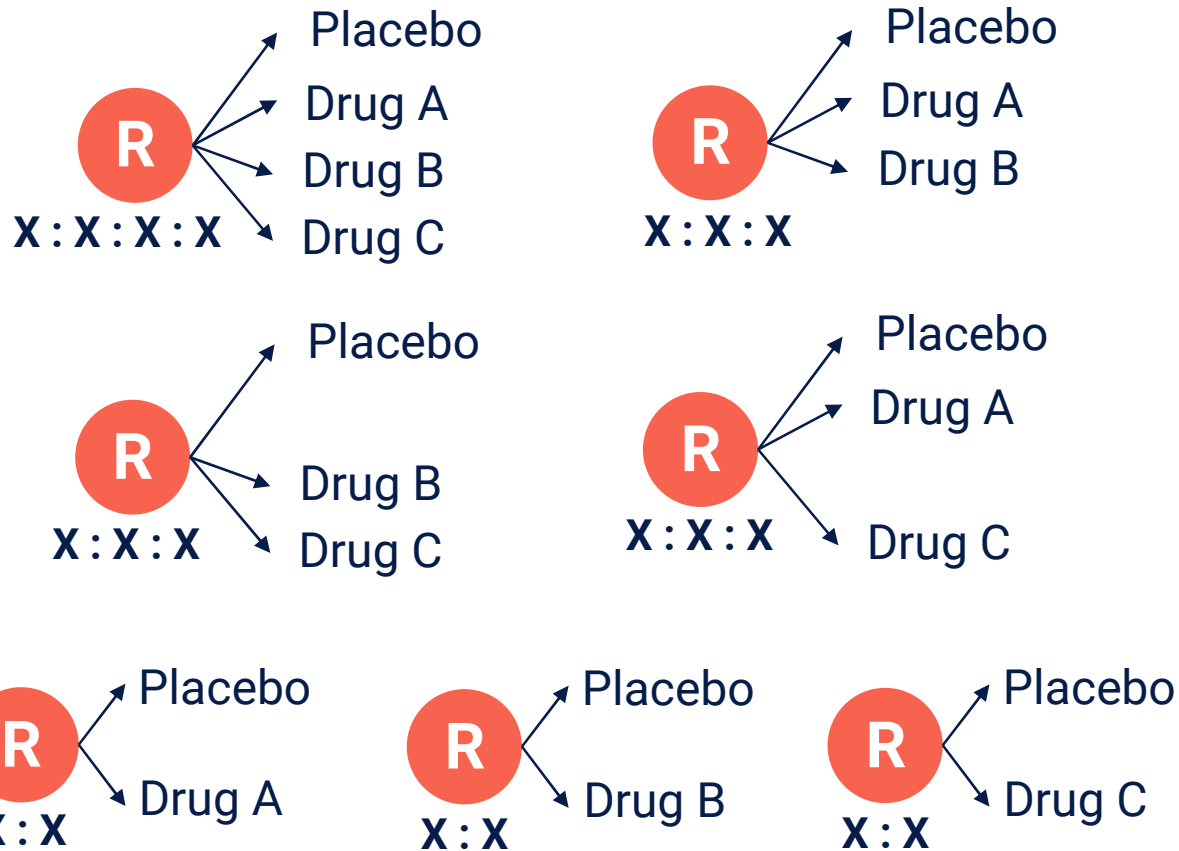
RN Schedule Strategy 2: One-Step RN Sets



This strategy accommodates

- discrete eligibility criteria across arms by predefining rules that assign subjects to their eligible RN schedule. E.g., subjects NOT eligible for Drug B will NOT be assigned to any RN schedules involving Drug B
- dynamic randomization ratio by setting allocation ratios within each RN schedule based on the #. of active arms
- May only enable full blinding – which requires triple-dummied doses when drugs A, B, and C have different schedules/routes

RN Schedule Strategy 2: One-Step RN Sets



Note of inferential impact:

- placebo within each RN schedule inherently serves as the valid concurrent control (naturally-justified as subjects being randomized within the same RN schedule)

RN Schedule Strategy 2: One-Step RN Sets

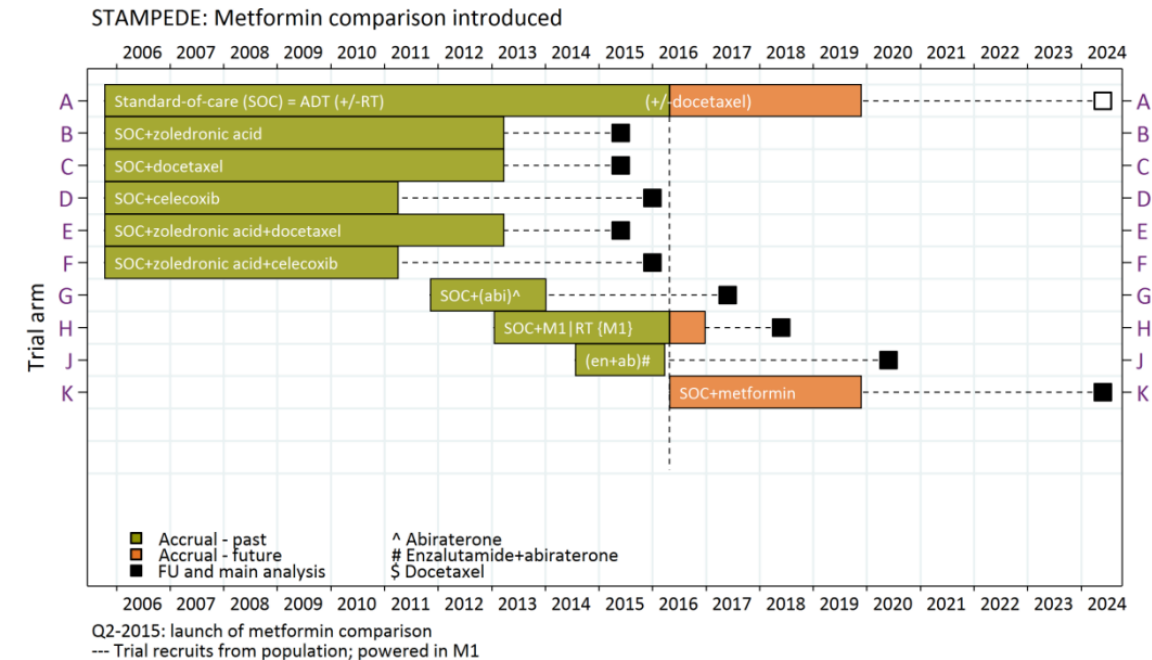
APPENDIX Q: HISTORY ALLOCATION AT RANDOMISATION BY PROTOCOL VERSION

Table 10: History of allocation at randomisation by protocol version

		NEWLY-DIAGNOSED M1 PATIENTS												OTHER PATIENTS																
PROTOCOL	PROTOCOL	NON-DIABETIC										DIABETIC		NON-DIABETIC										DIABETIC						
VERSION	CHANGE	A	B	C	D	E	F	G	H	J	K	A	B	C	D	E	F	G	H	J	K	A	B	C	D	E	F	G	H	J
V1	Not used	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-
V2	Activated	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-
V3		2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-
V4		2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-
V5		2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-
V6		2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-	-	2	1	1	1	1	1	-	-	-
V7	DF stopped	2	1	1	~	1	~	-	-	-	-	2	1	1	~	1	~	-	-	-	-	2	1	1	~	1	~	-	-	-
V8	G added	2	1	1	~	1	~	2	-	-	-	2	1	1	~	1	~	2	-	-	-	2	1	1	~	1	~	2	-	-
V9 ¹	H added	2	1	1	~	1	~	2	2	-	-	2	1	1	~	1	~	2	2	-	-	2	1	1	~	1	~	2	x	-
V10	BCE stopped	2	~	~	~	~	~	2	2	-	-	2	~	~	~	~	~	2	2	-	-	2	~	~	~	~	~	2	x	-
V11	G stopped	2	~	~	~	~	~	~	2	-	-	2	~	~	~	~	~	~	2	-	-	x	~	~	~	~	~	~	x	-
V12	J added	2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	x	2
V13		2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	x	2
V14	SOC change	2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	2	2	-	2	~	~	~	~	~	~	x	2
V15 ²	K added	2	~	~	~	~	~	~	2	~	2	2	~	~	~	~	~	~	2	~	x	2	2	~	~	~	~	~	x	~

1 or 2 Allocation weight while open to recruitment

Figure 2: Arms of the STAMPEDE trial open to recruitment over time



Case Example: Study STAMPEDE (Systemic Therapy in Advancing or Metastatic Prostate Cancer: Evaluation of Drug Efficacy).
ClinicalTrials.gov number, [NCT00268476](https://clinicaltrials.gov/ct2/show/study/NCT00268476),
and Current Controlled Trials number, [ISRCTN78818544](https://www.ccr-trials.com/record/ISRCTN78818544).

RN Schedule Strategy Evaluation

Design Feature	Two-Step RN with Updates Upon Arm Addition or Removal	One-Step RN Sets
Discrete eligibility criteria across arms	Handled via sub-study structure	Encoded in RN sets that reflect eligibility combinations
Dynamic Allocation Ratio	Updates to ratios required upon arm addition or removal	Pre-determined per-schedule ratios inherently adjust to the number of active arms
Pooling placebo data	Additional justification required	Naturally justified
Blinding Options	Partial blinding is feasible	Full blinding or Open-Label
Operational complexity	High (multi-step structure, updates needed per progress)	High (large number of schedules to set up)

Optimal Allocation Ratio in Platform Trial – Regulatory Guidance

VI. APPENDIX¹

A. Optimal Allocation Ratio

In certain scenarios, such as trials to evaluate more than two treatment groups compared to a common control, unequal allocation can improve efficiency (Chandereng et al. 2020). Here is a derivation in a simple case that power can be increased with disproportionately greater randomization to the control group in an umbrella or platform trial with a fixed total sample size. Consider an umbrella trial in which there are N total subjects, k drugs, and 1 control group. For some fraction p suppose that $N \times p$ subjects are assigned to control and $N \times (1 - p)/k$ subjects are assigned to each drug. Also, suppose that the treatment effect δ is the same for each drug and that outcomes for all groups have the same variance. The power of a z-test is determined by δ/σ , where σ^2 is the variance of the (treatment – control) difference in means and is proportional to $f(p) = 1/[N \times (1 - p)/k] + 1/(N \times p)$. Considering p as continuous in $(0, 1)$ (even though strictly speaking the number of treatment and control subjects should be integers) the first derivative of $f(p)$ is $1/N \times [k/(1 - p)^2 - 1/p^2]$. The second derivative of $f(p)$ is $1/N \times [2 \times k/(1 - p)^3 + 2/p^3] > 0$, so the function $f(p)$ is convex, and thus variance is minimized by setting the first derivative to zero. This is achieved at $p = 1/(1 + \sqrt{k})$ which is equivalent to a randomization ratio for the control relative to a given drug of $\sqrt{k}:1$. In contrast, equal allocation to all treatment groups would correspond to $p = 1/(1 + k)$. This example illustrates a simple case, and the Chandereng et al. 2020 paper shows more generally, that the optimal allocation will have disproportionate randomization to the control group when $k > 1$.

Master Protocols for Drug and Biological Product Development Guidance for Industry

DRAFT GUIDANCE

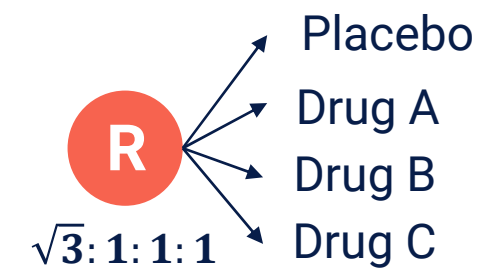
This guidance document is being distributed for comment purposes only.

Comments and suggestions regarding this draft document should be submitted within 60 days of publication in the *Federal Register* of the notice announcing the availability of the draft guidance. Submit electronic comments to <https://www.regulations.gov>. Submit written comments to the Dockets Management Staff (HFA-305), Food and Drug Administration, 5630 Fishers Lane, Rm. 1061, Rockville, MD 20852. All comments should be identified with the docket number listed in the notice of availability that publishes in the *Federal Register*.

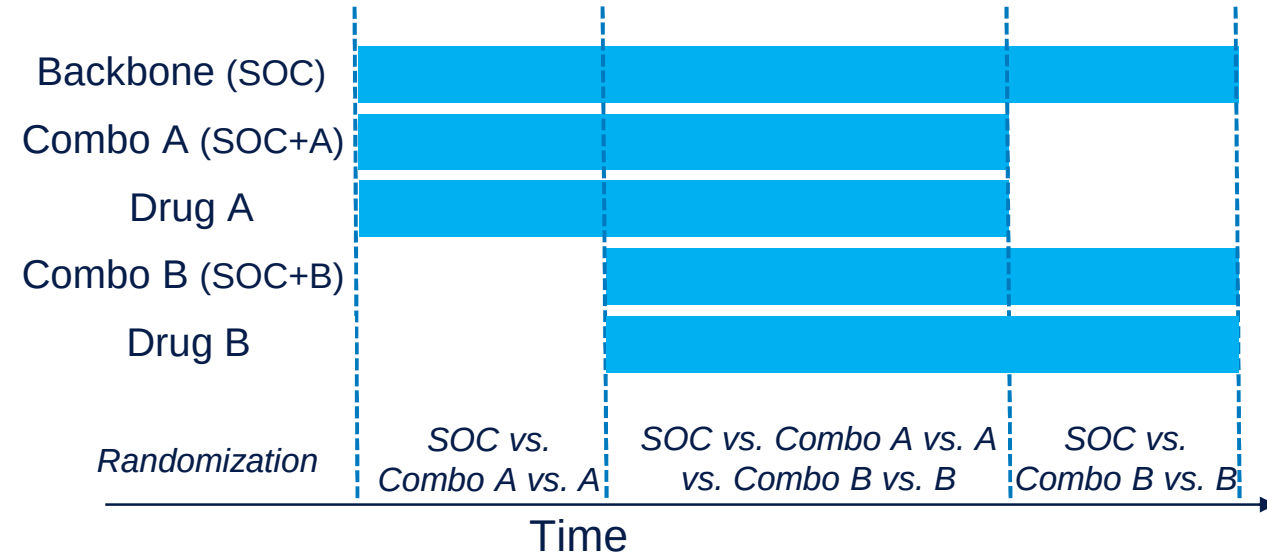
For questions regarding this draft document, contact (CDER) Scott N. Goldie at 301-796-2055, or (CBER) Office of Communication, Outreach and Development, 800-835-4709 or 240-402-8010.

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)

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Biostatistics / Clinical / Medical



Case Study: Combination Therapy Evaluation in Platform Trial



Combination Therapy

- May offer superior efficacy, addressing unmet medical needs
- FDA requires demonstrating the contribution of each therapy component

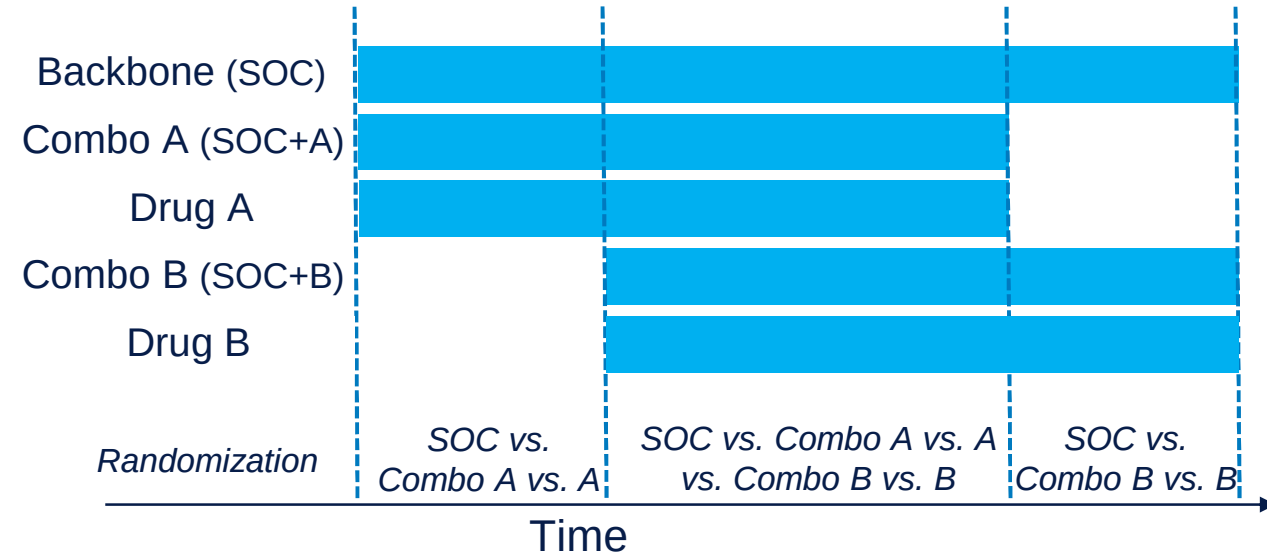
Statistical Challenges in Randomization Ratio

- Backbone comparator and combo arms will be compared multiple times – *is it still max. power when assigning more subjects to the backbone only?*
- Timing of adding or discontinuing arms – *Considering the potential staggered availability of investigational drugs, how do we choose the optimal timing? how would it impact randomization over time?*

Research Question

Objective: optimizing allocation ratios, considering

- fix total sample size N
- pairwise comparisons
 - Combo X vs Backbone; Combo X vs Drug X (X=A,B)
 - **concurrent** comparison
- optimization goal
 - to maximize the minimum power across investigational treatments
 - to ensure adequate power for the worst-performing test

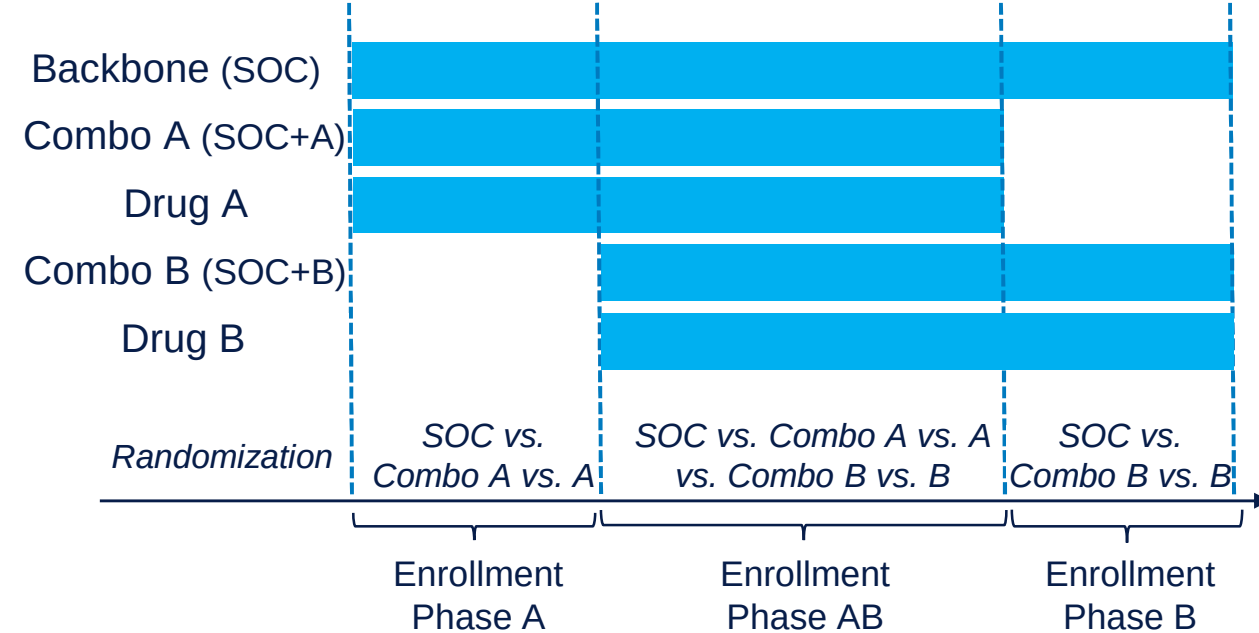


Research Question

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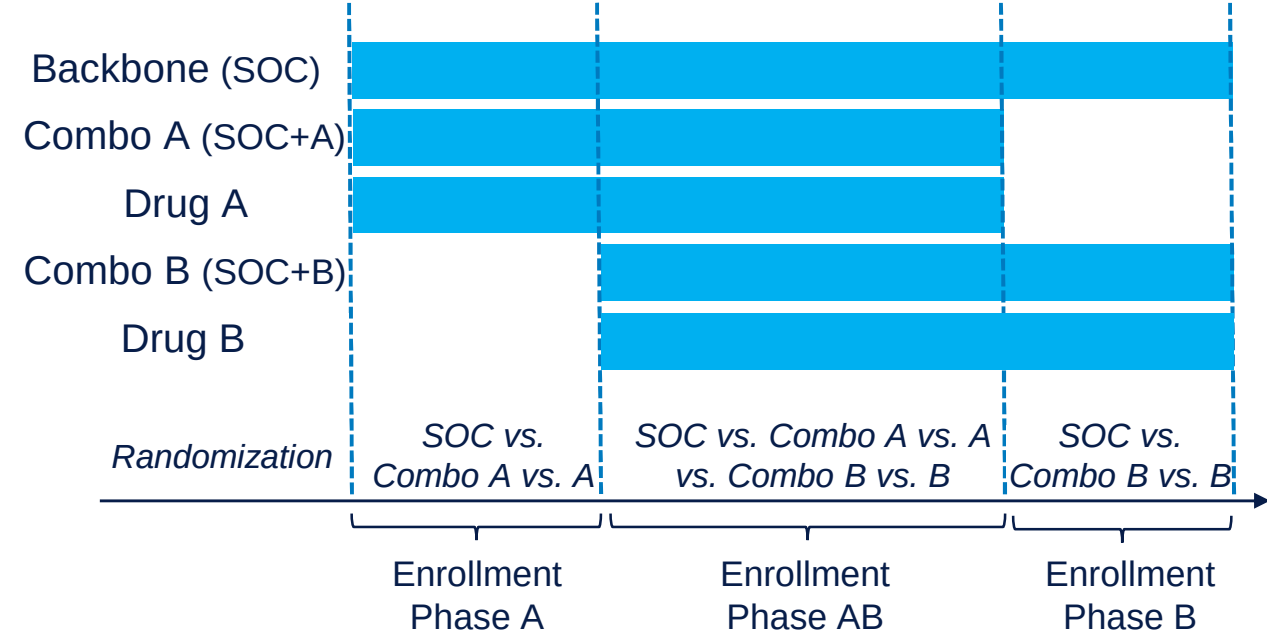
- the number of arms and their entry and exit time
 - stratified testing estimator, adjusting for changing ratios per enrollment phase, to formulate marginal power (in line with FDA guidance to avoid bias)

Approach: theoretical derivation & numerical optimization



Simulation Setup

- Total sample size N=200
- Endpoint follows a Normal Distribution
 - $Y_i \sim N(\mu_i, \sigma^2 = 1)$
 - $\mu_i = \begin{cases} 0, & i = \{Backbone, Drug A, Drug B\} \\ 0.5, & i = \{Combo A, Combo B\} \end{cases}$
- Allocation ratios under evaluation:
 - derived optimal allocation
 - equal allocation
- Simulated 5000 trials to obtain empirical power for each test



Design Scenario: Staggered Launch & Flexible Exit

- Restriction: Combo Set B is scheduled to be launched later than Combo Set A; but no restrictions on when arms must be dropped
- *Fixed design factors*: total sample size, timing of adding Combo Sets A/B
- *Design factors to be determined*: timing of dropping Combo Sets A/B, corresponding randomization ratio by enrollment phase

Sub-scenario 1 if, by the time Combo Set B is added, the accumulated sample size has reached half of the total sample size

✓Optimal Timing:

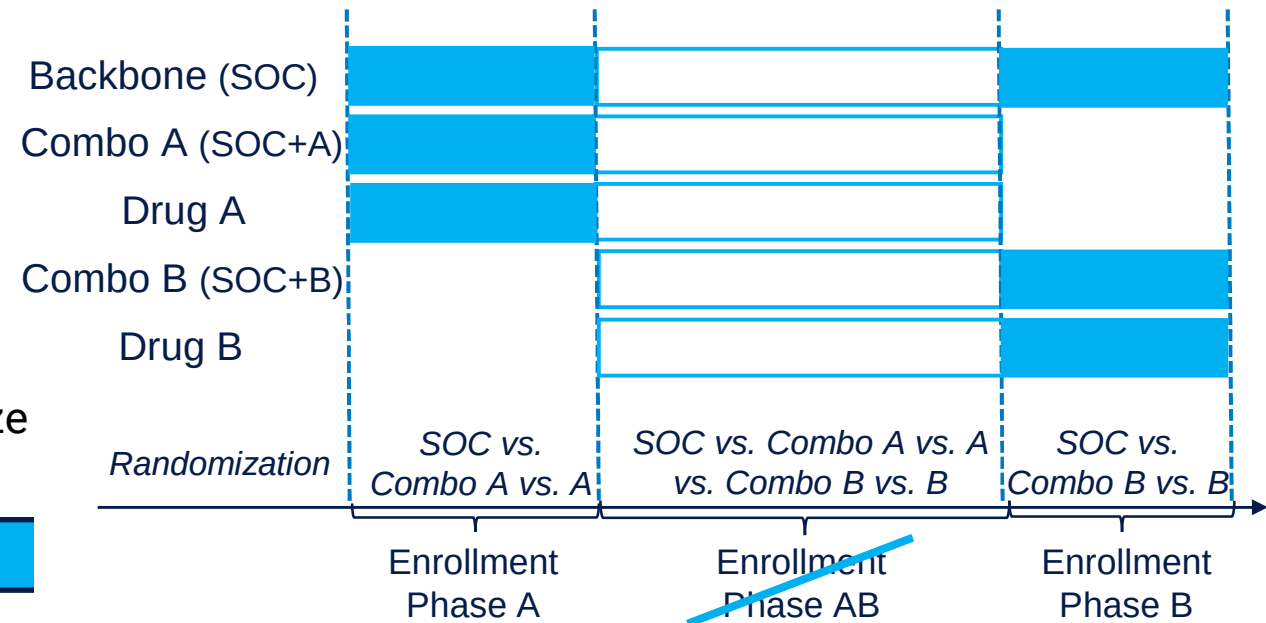
- evaluate two combo sets separately

✓Optimal Randomization:

- assign more subjects to combo arms
- 1 Backbone vs. $\sqrt{2}$ Combo A/B vs. 1 A/B

✓**Power Gain** (a function of the proportion of sample size allocated to Combo Set A evaluation):

	0.55	0.6	0.65	0.7
Optimal	0.492	0.458	0.402	0.363
Equal	0.489	0.444	0.398	0.346



Design Scenario: Staggered Launch & Flexible Exit

- Restriction: Combo Set B is scheduled to be launched later than Combo Set A; but no restrictions on when arms must be dropped
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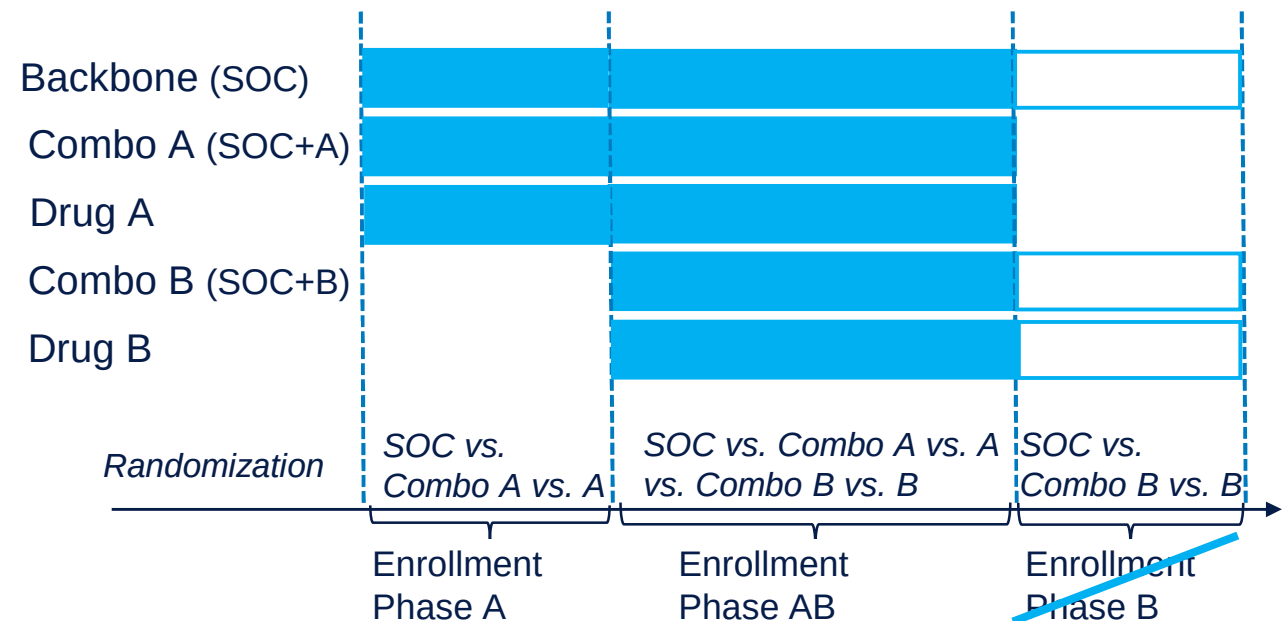
Sub-scenario 2 if, by the time Combo Set B is added, the accumulated sample size has NOT reached half of the total sample size

✓ **Optimal Timing:**

- end two combo sets at the same time

✓ **Optimal Randomization & Power Gain:**

- dependent on the proportion of sample size allocated to Combo Set A evaluation before adding Combo Set B (i.e., Enrollment Phase A)



Design Scenario: Staggered Launch & Flexible Exit

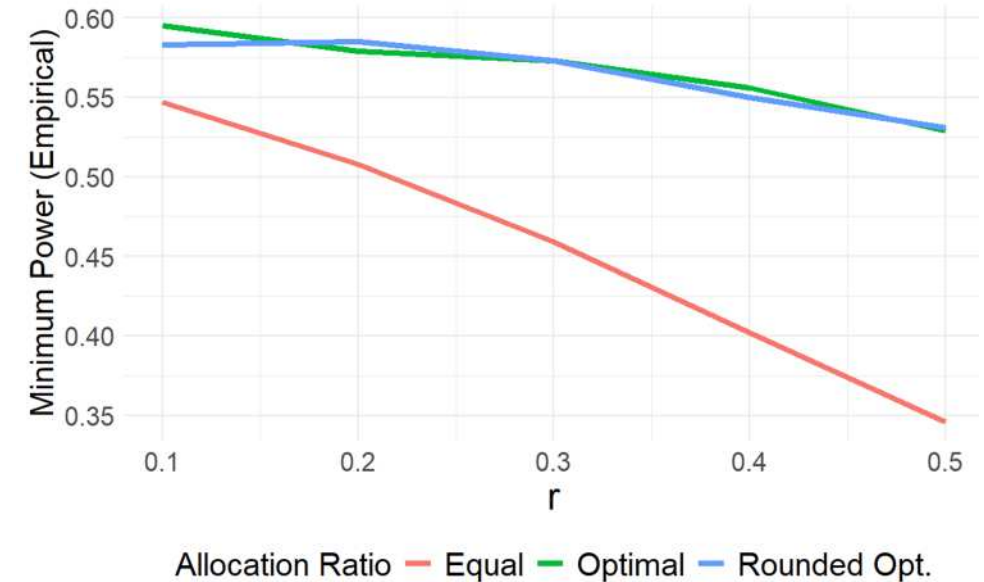
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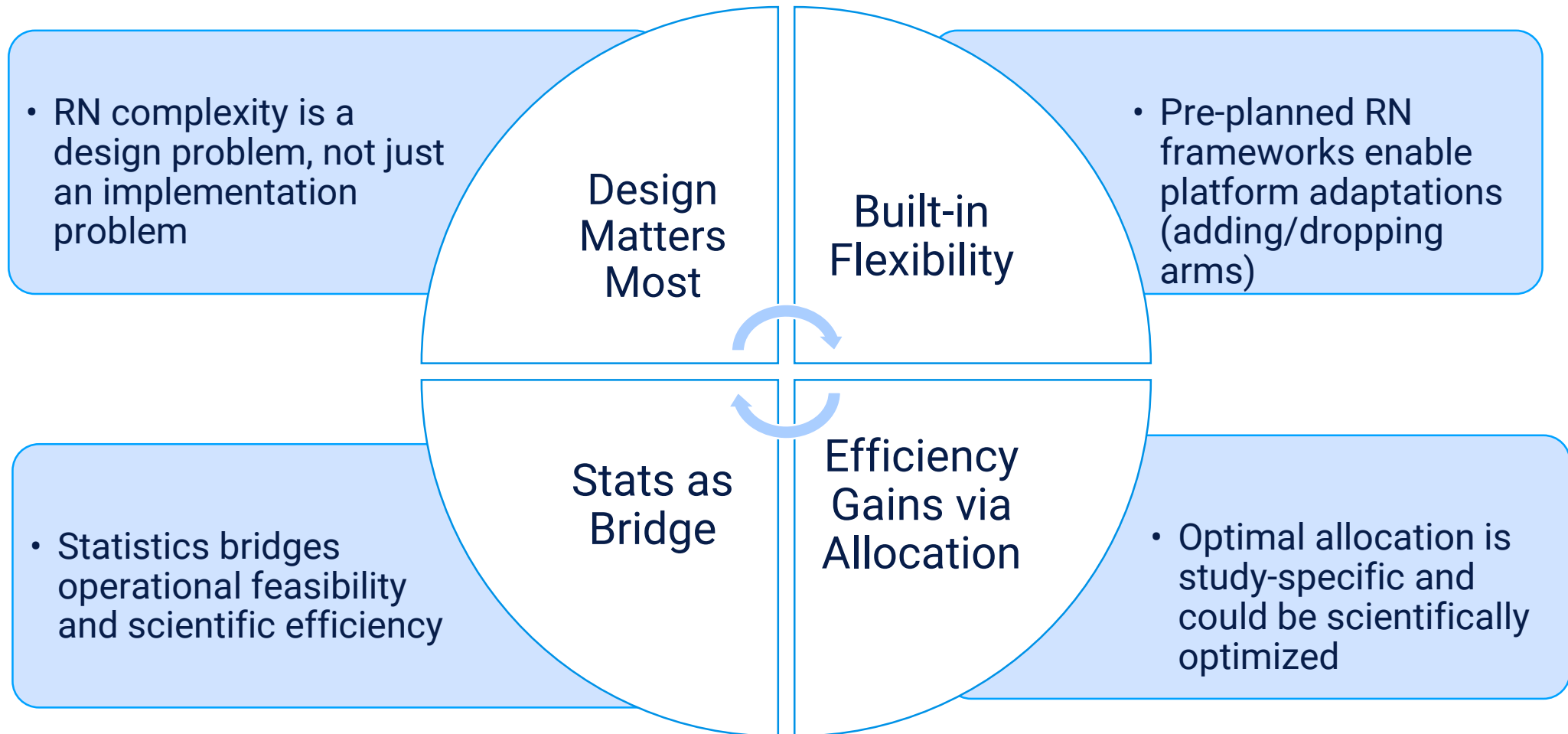
✓ **Power Gain** (a function of the r : proportion of sample size allocated to Combo Set A evaluation before adding Combo Set B)

Allocation Ratio	r				
	0.1	0.2	0.3	0.4	0.5
Equal	0.547	0.508	0.459	0.402	0.346
Optimal	0.595	0.579	0.573	0.556	0.529
Rounded Opt.	0.583	0.585	0.573	0.550	0.531

Allocation Ratio	$p_A^{SOC} : p_A^{Combo A} : p_A^{Drug A}$			$p_{AB}^{SOC} : p_{AB}^{Combo A} : p_{AB}^{Drug A} : p_{AB}^{Combo B} : p_{AB}^{Drug B}$				
Equal	1	1	1	1	1	1	1	1
Optimal	1	2.8	2.6	1.8	1.4	1	2.3	1.8
Rounded Opt.	2	6	5	4	3	2	5	4



Discussion



Platform trials introduce operational and inferential complexity – but it is solvable with principled statistical design.

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